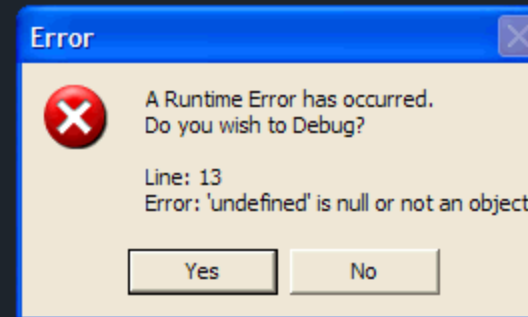


OPENVALUE
TECH TRIBES

Beyond `system.out.println`: Applying a Debugging Mindset at Scale

Debugging Mindset at Scale

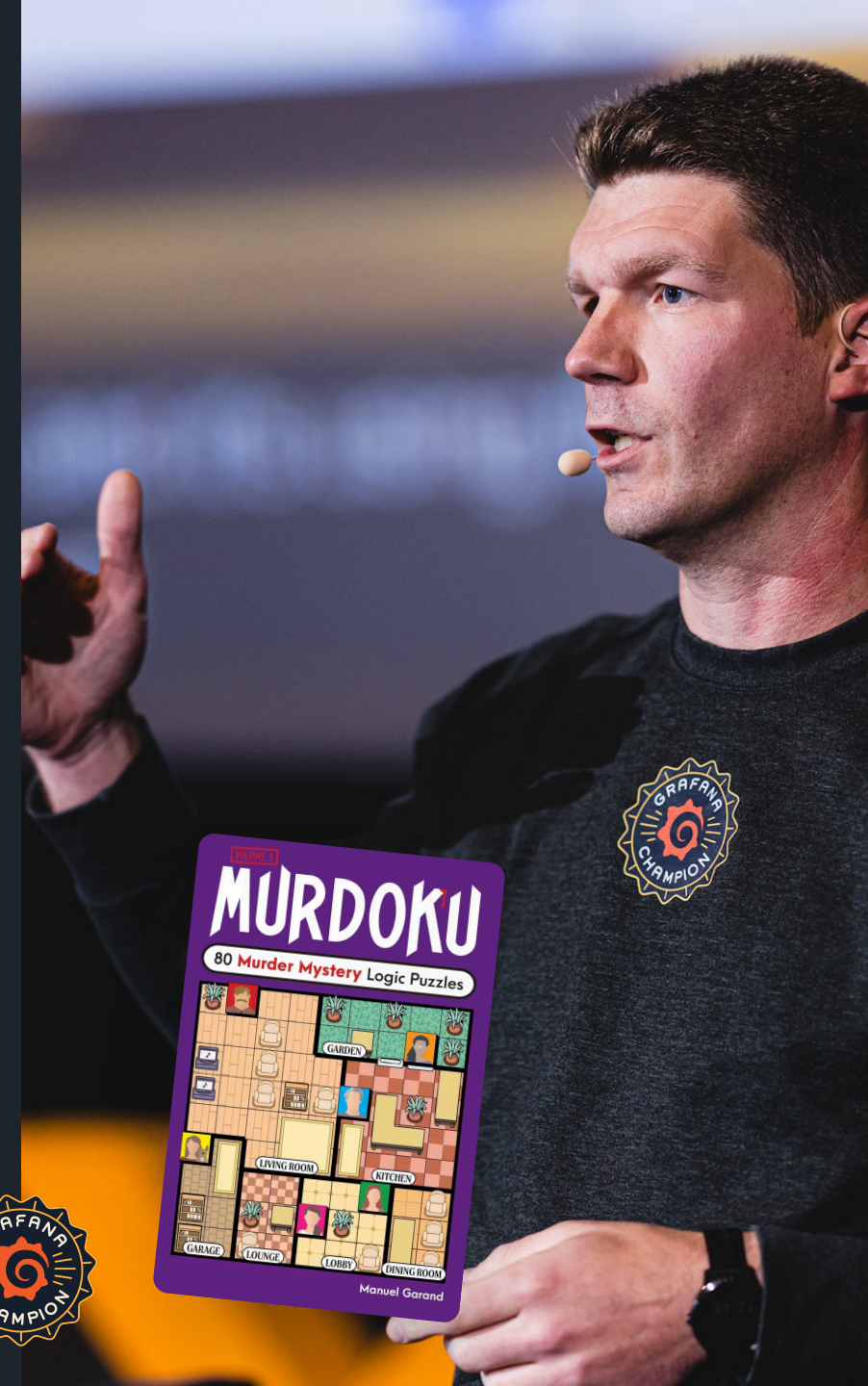
```
debugger;
```



About me - Cees Bos

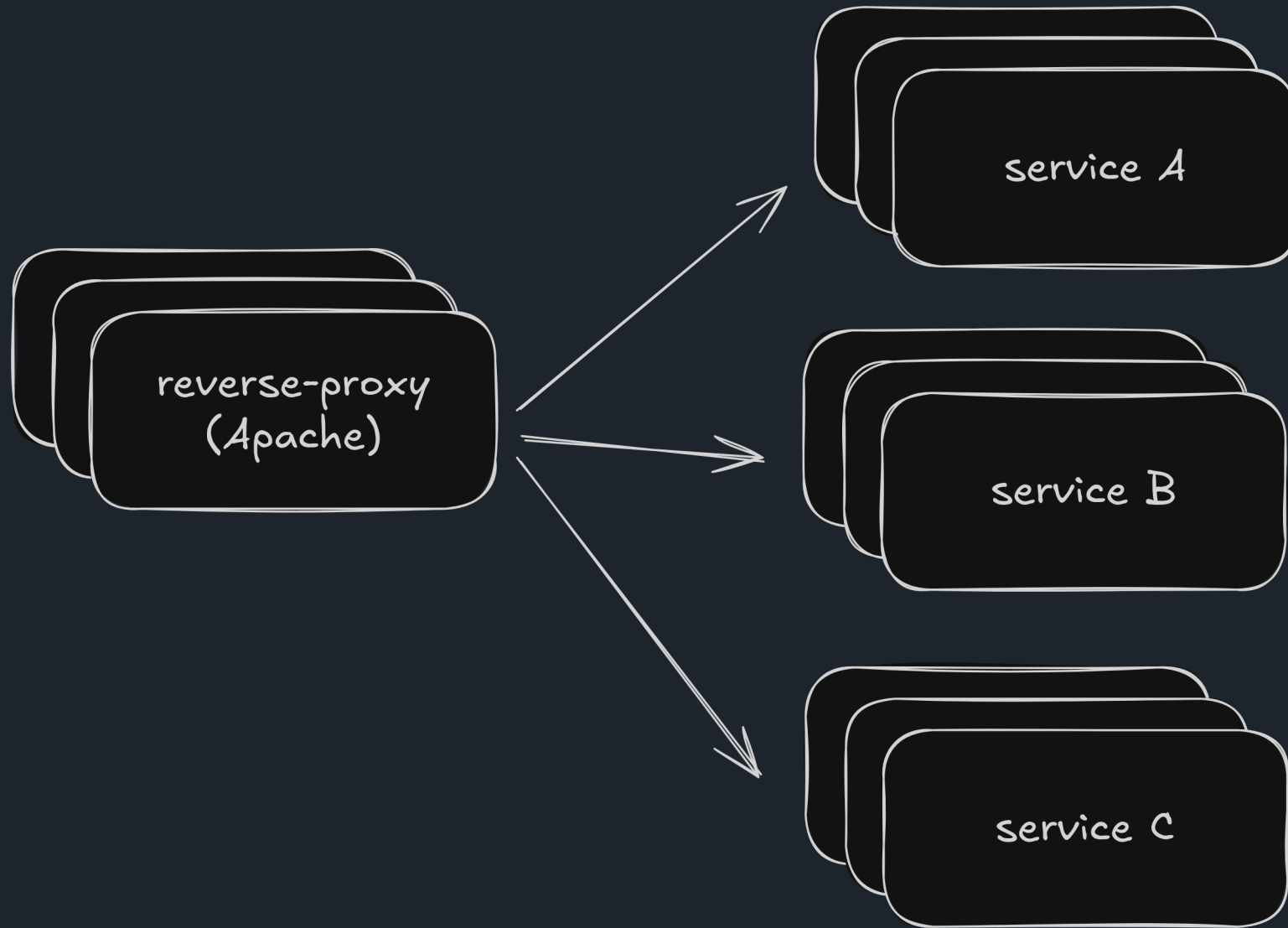
- Software & Observability Engineer and SRE at OpenValue
 - Grafana Champion
-
- Always interested in how an application is actually doing on production and how to improve that
 - Observability enthusiast always looking for more insights
-

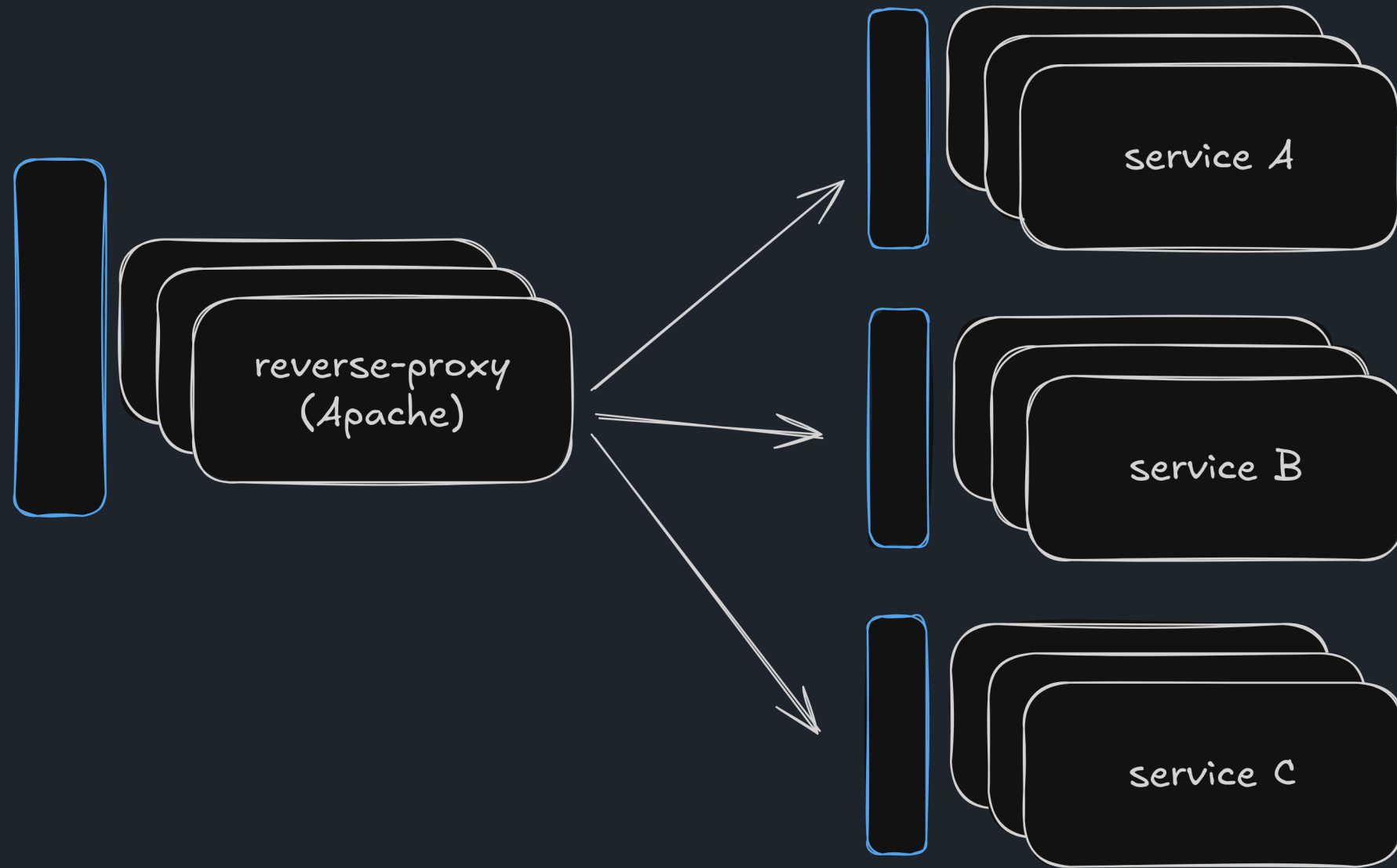
<https://ceesbos.nl> and <https://openvalue.blog/>

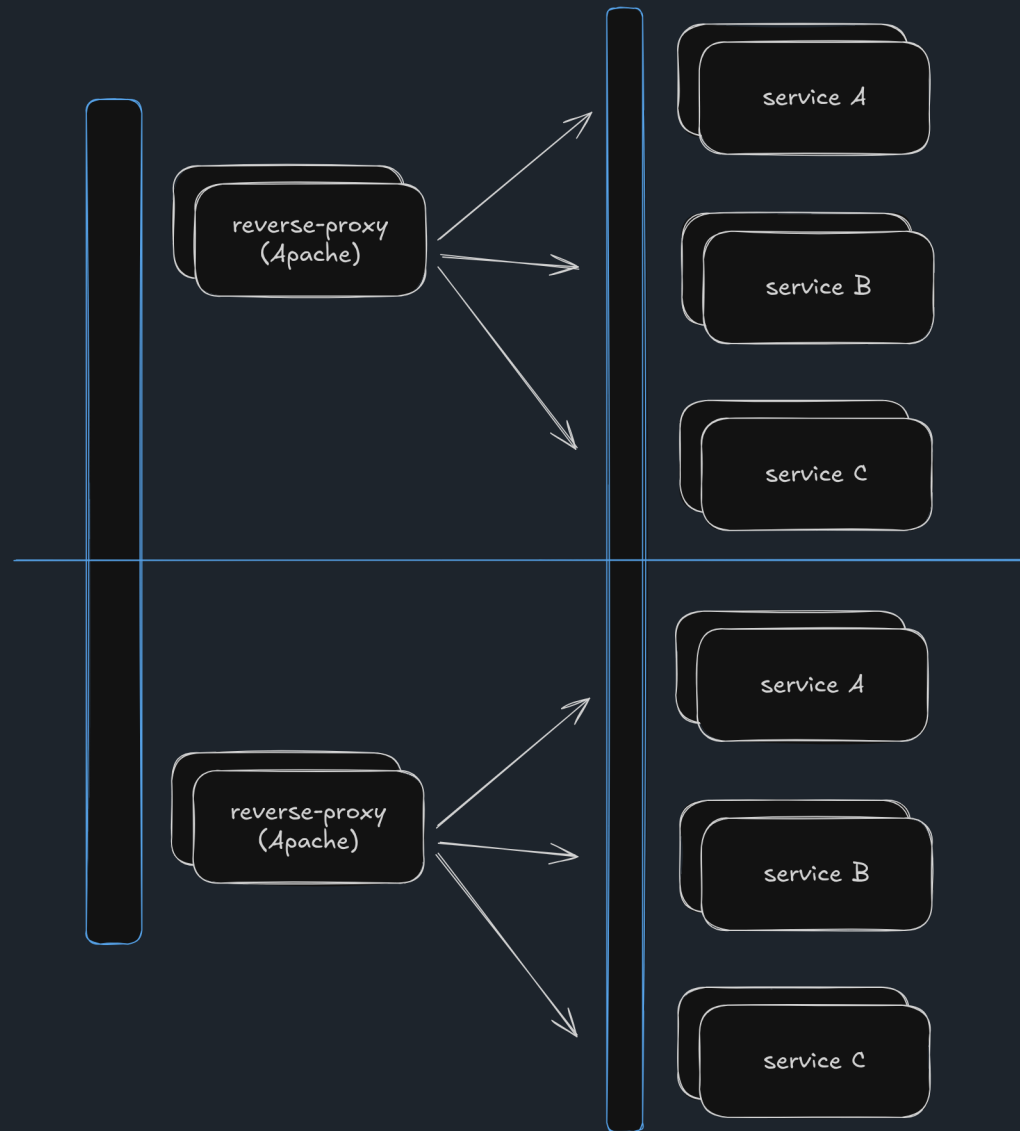


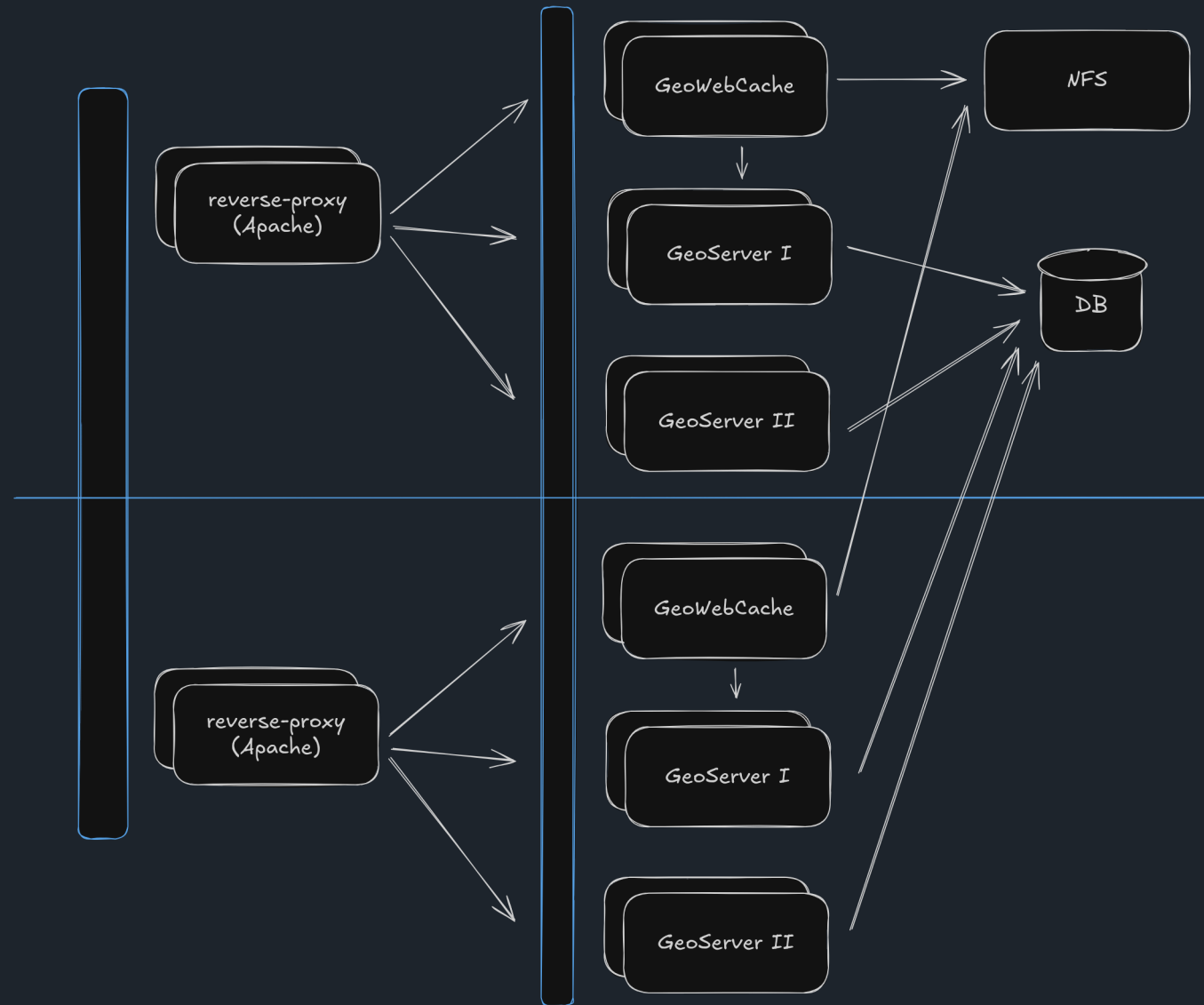
What can you expect during this talk

- The landscape
- The issues
- The approach
- The lessons learned





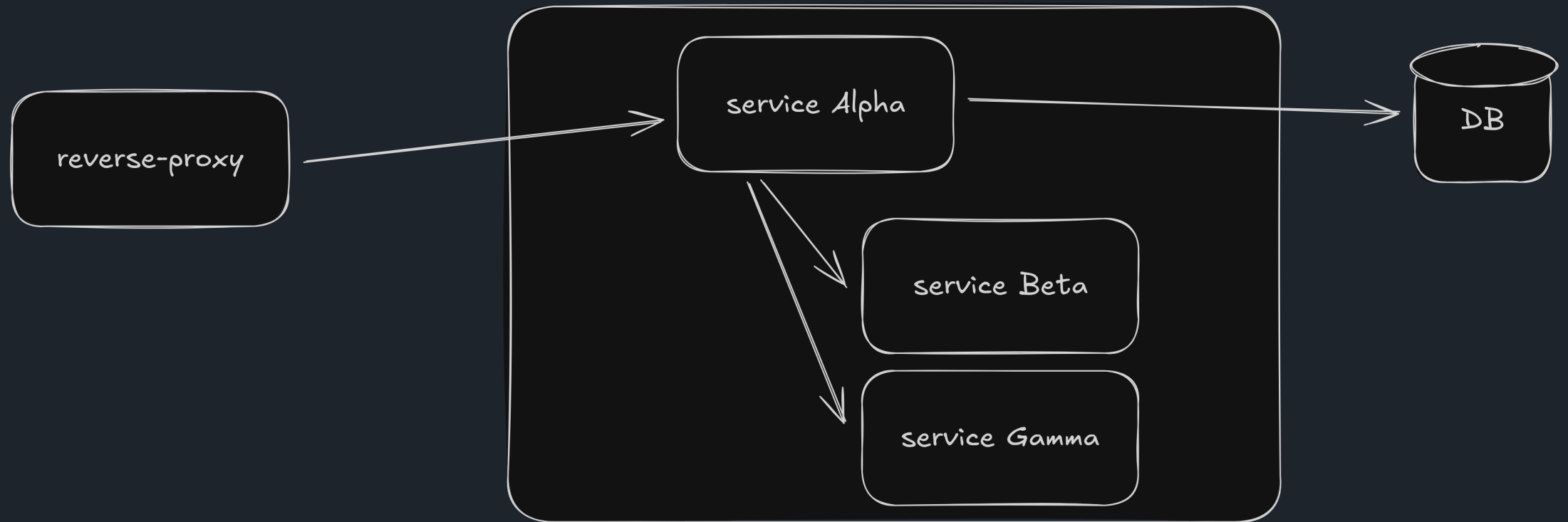




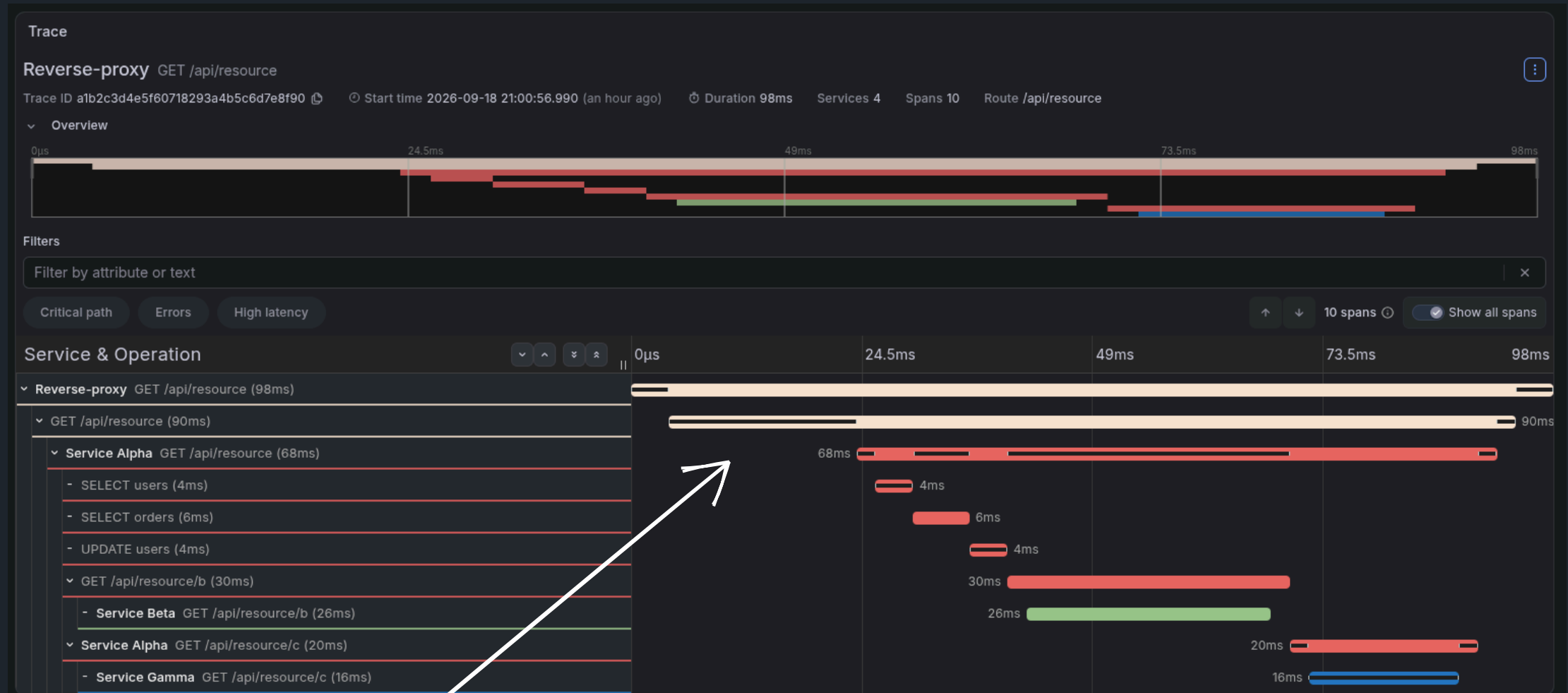
Lessons

- Make sure you know the surrounding environment
- Keep asking questions
- Don't let it go
- Address issues faced with other components
- Fill the gaps of your observability

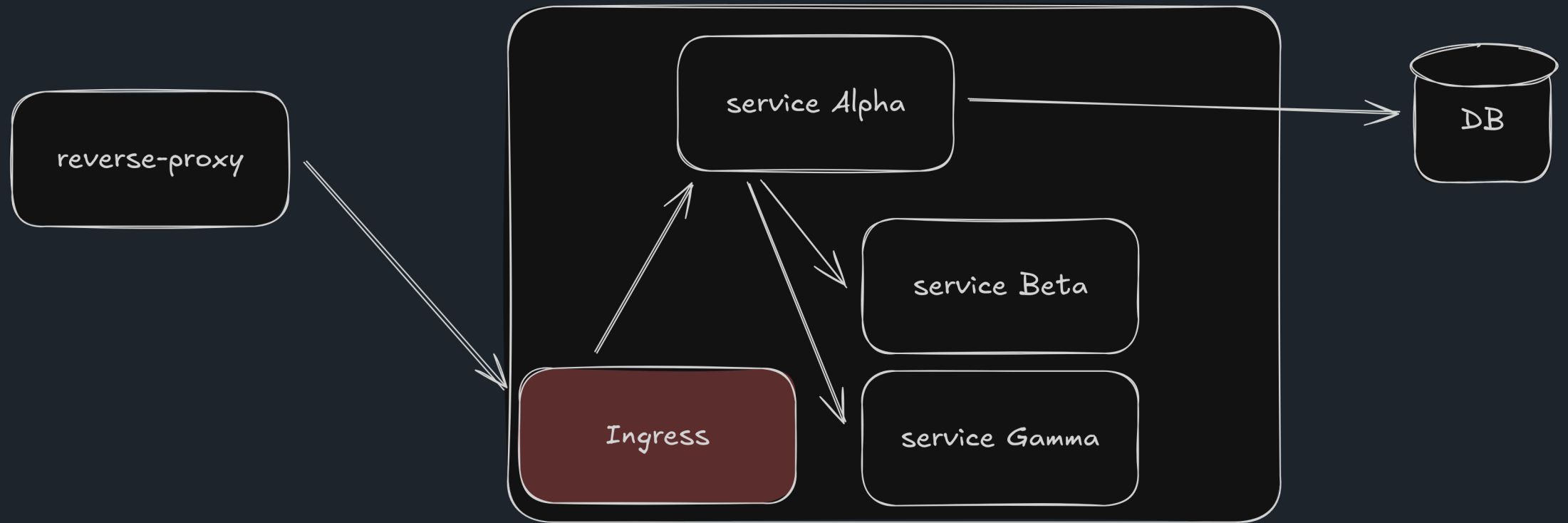
Kubernetes



Kubernetes



Kubernetes



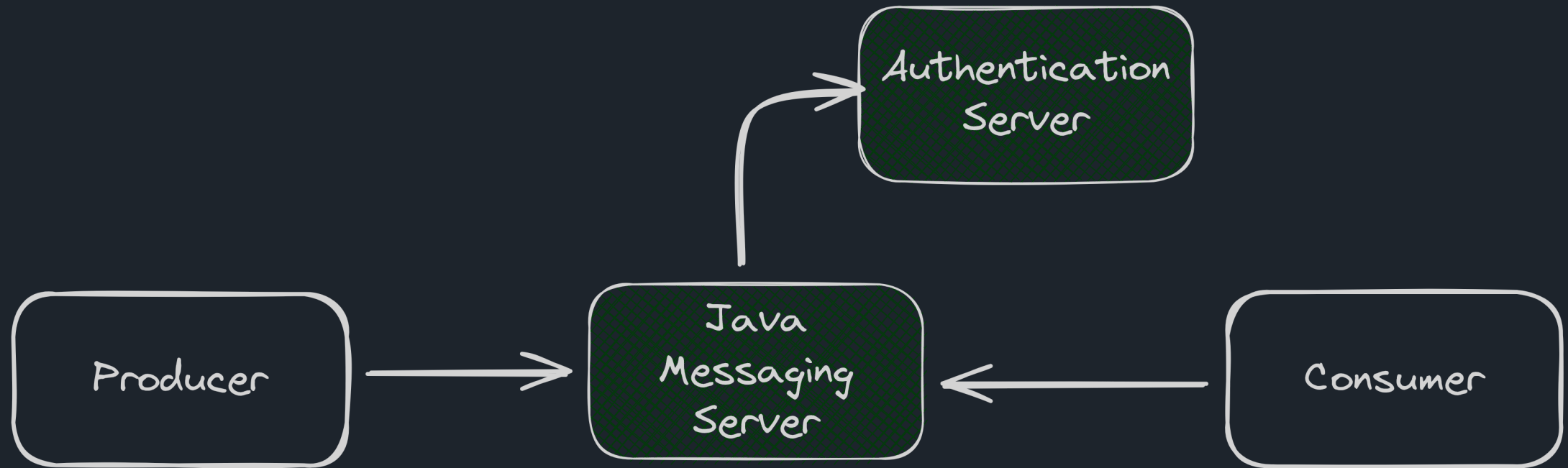
Lessons

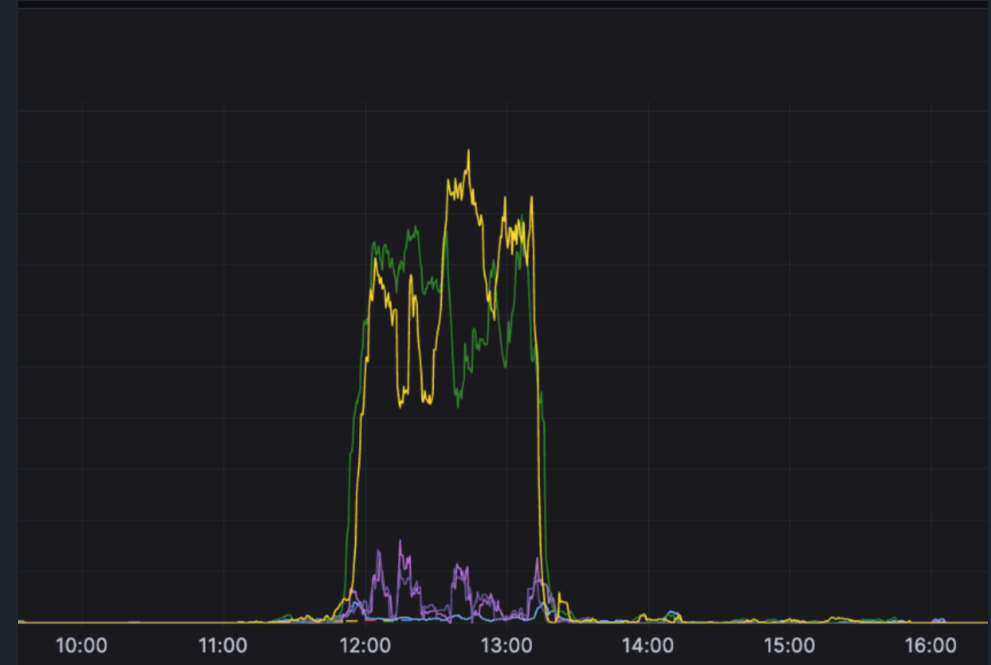
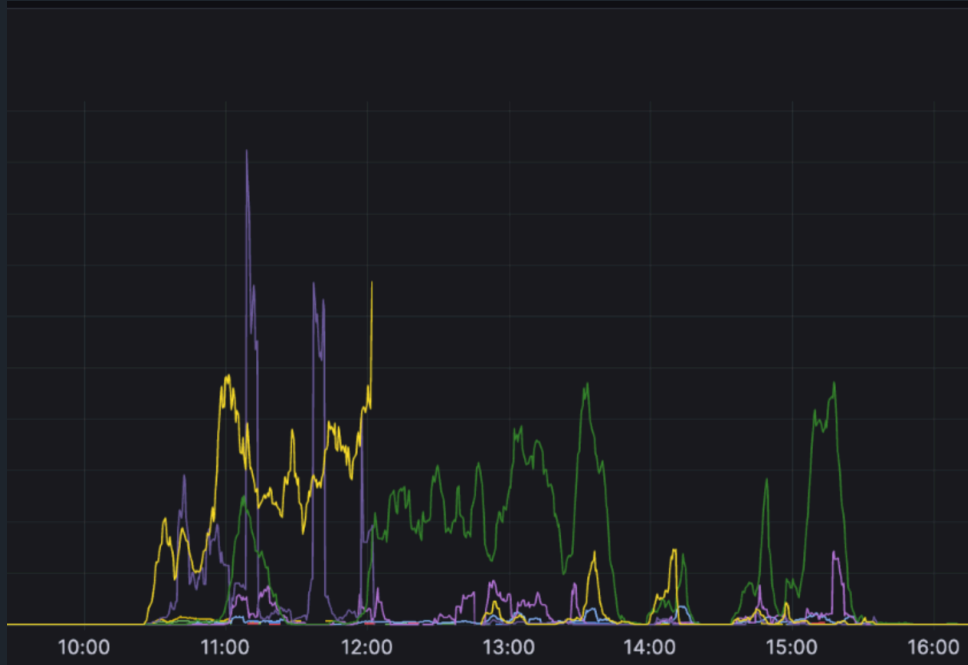
- Make sure you know the surrounding environment and how it works under the hood
- Keep asking questions
- Don't let it go
- Address issues faced with other components
- Fill the gaps of your observability

Large landscape impact



Large landscape impact





"Assume it is you, prove it is not you"

Make a hypothesis and either prove or disprove it

- Observability data (with OpenTelemetry)
 - Metrics
 - Logs
 - Traces
 - Profiling
- Observability platform (like Grafana)

Observable?

- How *effectively* you can understand system behaviour from the outside using the data it generates.



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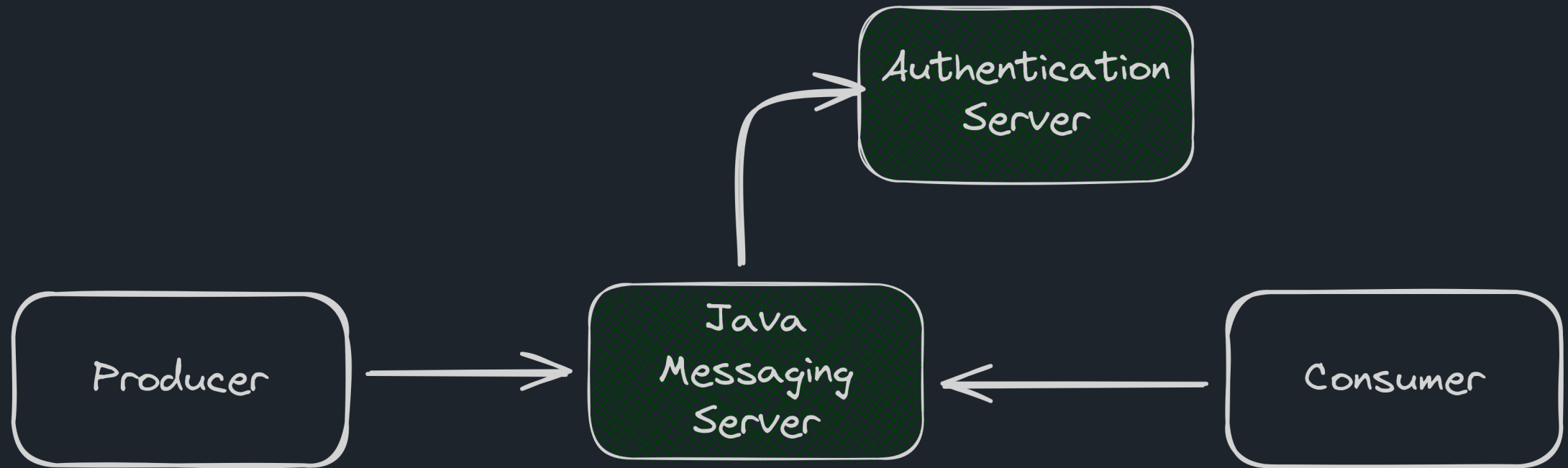
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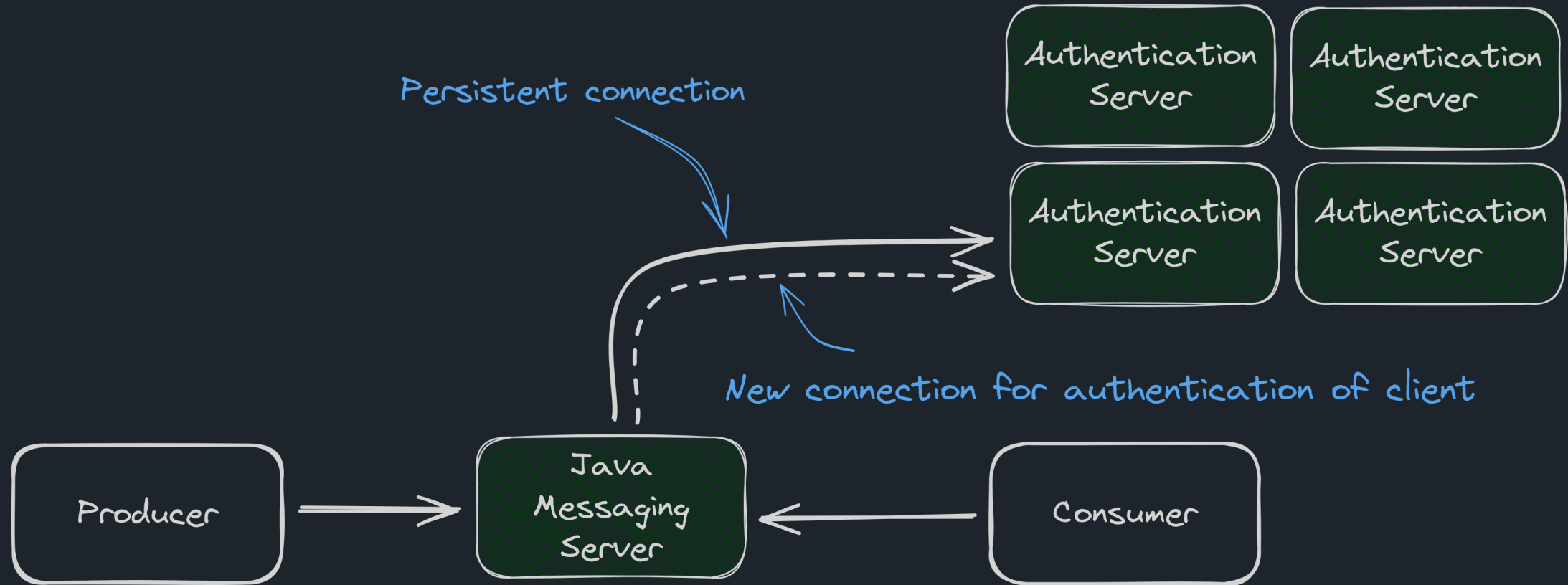
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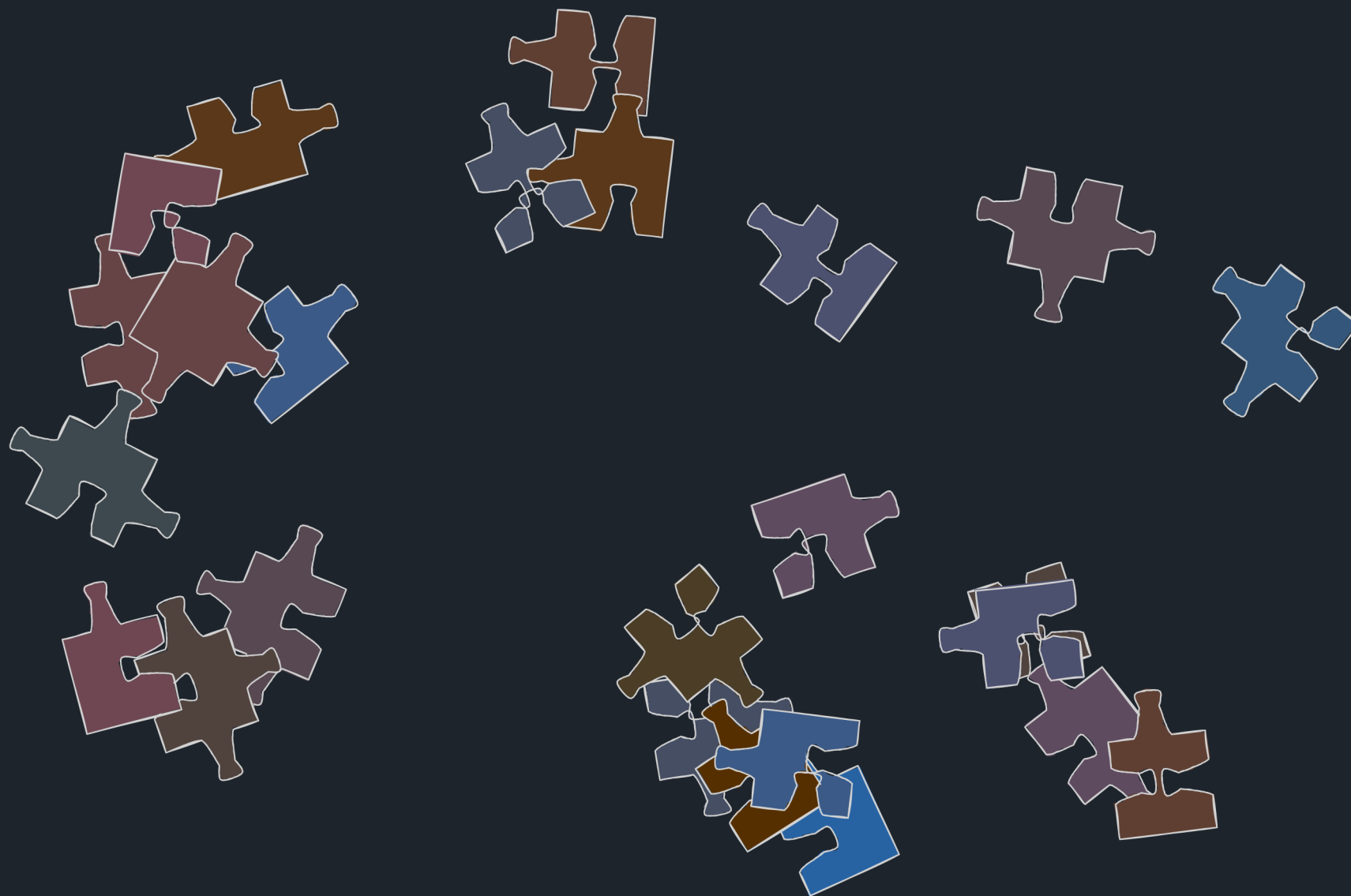
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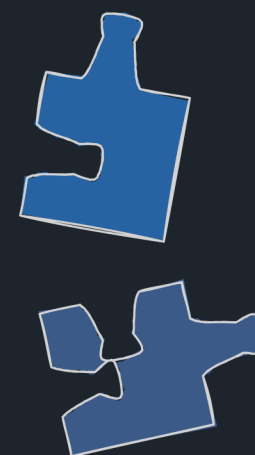
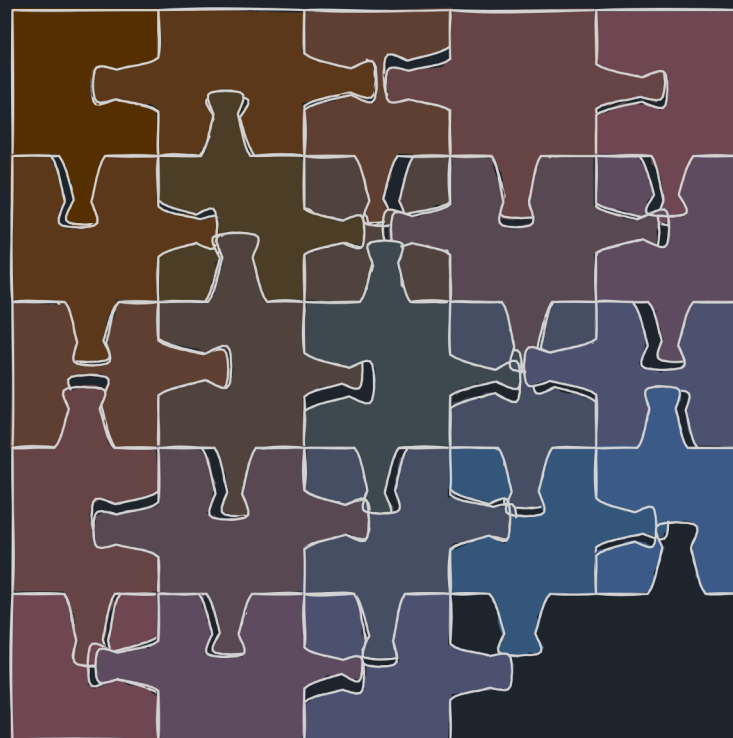
Large landscape impact



Large landscape impact







Data Driven Analysis with visualizations

- Drawings
- Visualize the observability data
- Timelines
- Share the analysis

Lessons

- Make sure you know the surrounding environment and how it works under the hood
- Keep asking questions
- Don't let it go
- Address issues faced with other components
- Don't make assumptions; make a hypothesis and prove it
- Zoom in into a specific part and figure out all details
- Complete the whole puzzle, don't leave unanswered questions open
- Foundation: (Observability) Data + curiosity + perseverance

Questions?

Thank for your time!

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DON'T FORGET TO RATE THIS TALK!